Voice Assistance for Multimedia Control

Hardware-Based Voice Control System using Raspberry Pi / ESP32

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2. Software Test Plan

1. Introduction

1.1 Purpose

This Test Plan describes the scope, approach, resources, and schedule for testing the Voice Assistance for Multimedia Control System. It ensures that every functional and non-functional requirement stated in the Software Requirements Specification (SRS) is verified before the system is accepted as complete.

1.2 Scope

Testing covers voice-command recognition and execution of all seven multimedia control actions (Play, Pause, Resume, Next, Previous, Volume Up, Volume Down), along with the supporting non-functional qualities (response time, reliability, availability) and the security controls defined in the SRS (wake-word gating and local processing). Testing is performed on the target hardware (Raspberry Pi / ESP32) running the final integrated build.

1.3 References

- IEEE Std 829-2008, IEEE Standard for Software and System Test Documentation.
- Software Requirements Specification (SRS) – Voice Assistance for Multimedia Control, v1.0.
- Project handout: SE Mini-Project Deliverables Part-1.

2. Test Items

The following items are subject to testing in this plan:

- Wake-word / voice command listener module (microphone input handling).
- Speech-to-command recognition module (ASR + command mapping).
- Command execution module (interface to the media player: Play, Pause, Resume, Next, Previous, Volume Up, Volume Down).
- Feedback module (LED / log acknowledgement on command execution).
- Security module (wake-word gating and local-only audio processing).
- Integrated build running on Raspberry Pi and/or ESP32 target hardware.

3. Features to be Tested

All features traceable to the requirements in the SRS are in scope, specifically:

- Wake-word detection and listening state (FR-1).
- Execution of each of the seven voice commands (FR-2 to FR-8).
- Rejection of unrecognized / out-of-vocabulary audio input (FR-9).
- Execution of feedback / acknowledgement (FR-10).
- Command response time (NFR-1) and recognition accuracy (NFR-2).
- Continuous availability over an extended listening session (NFR-3).
- Security controls: wake-word gating (SR-1) and local processing (SR-2).

4. Features Not to be Tested

The following are explicitly out of scope for this test cycle, consistent with the SRS constraints:

- Free-form conversational / general-purpose AI interaction (not part of the defined command vocabulary).
- Multi-user concurrent voice commands.
- Internet search, smart-home control, or any function beyond multimedia control.
- Third-party media player internals (only the control interface exposed to this system is tested).

5. Approach

Testing follows a bottom-up strategy: individual modules (wake-word detection, ASR, command mapping, execution, feedback) are unit tested first, followed by integration testing of the full voice-to-action pipeline, and finally system-level testing on actual Pi/ESP hardware in a realistic usage environment. Both black-box testing (for functional command behavior) and measurement-based testing (for timing and accuracy of NFRs) are used. Each test case is mapped to a specific requirement of ID for traceability (see Section 13).

5.1 Security Validation Approach

Security validation focuses on verifying the two security requirements defined in the SRS:

- Wake-word gating (SR-1) is validated by playing background/ambient audio and unrelated speech that does NOT contain the wake word and confirming that no command is executed.
- Local processing (SR-2) is validated by monitoring network traffic during command execution to confirm that raw audio is not transmitted off-device during normal operation.

Security test results are recorded using the same test case format as functional tests (see TC-10 in Section 14) and are reviewed separately by the team before signing-off, given their importance to overall system trustworthiness.

6. Item Pass/Fail Criteria

- A test case passes if the actual result matches the expected result stated for that test case, within any stated tolerance (e.g. timing thresholds).
- A feature is considered verified when all test cases are mapped to its requirement ID to pass.
- The overall test cycle is considered successful when at least 90% of all test cases pass, with zero failures on any security test case (SR-1, SR-2).

7. Suspension Criteria and Resumption Requirements

- Testing is suspended if the hardware (Pi/ESP) or microphone becomes non-functional, or if a build fails to install/boot on the target device.
- Testing resumes once the blocking hardware or build issue is resolved and confirmed working by a quick smoke test (one Play and one Pause command).

8. Test Deliverables

- This Test Plan document.
- Test case log with actual results (Section 14, updated after execution).
- Traceability matrix (Section 13).
- Defect / issue log for any failed test cases.
- Final test summary report.

9. Test Environment

- Hardware: Raspberry Pi 4 / ESP32 development board, USB or I2S microphone, speaker/media output device.
- Software: Python 3.x, speech recognition library (e.g. Speech Recognition / MediaPipe), target media player application.
- Test setting: quiet indoor room for baseline tests; controlled background-noise setting (e.g. playing music at low volume) for robustness and security tests.

10. Responsibilities

Role	Responsibility
Project Team	Design and execute all test cases; record actual results; log defects.
Team Lead	Review test results, prioritize defect fixes, sign off on pass/fail status.
Course Evaluator	Review test plan and results as part of the mini-project deliverable assessment.

11. Schedule

Activity	Start	End
Unit testing of modules	Day 1	Day 2
Integration testing (voice-to-action pipeline)	Day 3	Day 3
System testing on target hardware	Day 4	Day 4
Security validation testing	Day 4	Day 5
Test summary and sign-off	Day 5	Day 5

12. Risks and Contingencies

Risk	Impact	Contingency
Background noise causes false command recognition	Medium	Add noise-filtering steps; retest in a quieter environment.
Hardware (Pi/ESP) unavailable or faulty during test window	High	Keep a backup device configured and ready.
ASR library differs in accuracy between Pi and ESP32	Medium	Test each target separately; document accuracy per platform.

13. Traceability Matrix (SRS to Test Cases)

The table below maps each SRS requirement to the test case that verifies it, ensuring full traceability between requirements and testing as required for this deliverable.

Req. ID	Requirement	Test Case ID
FR-1	Wake / Listen	TC-01
FR-2	Play Command	TC-02
FR-3	Pause Command	TC-03
FR-4	Resume Command	TC-04
FR-5	Next Track Command	TC-05
FR-6	Previous Track Command	TC-06

FR-7	Volume Up Command	TC-07
FR-8	Volume Down Command	TC-08
NFR-1	Performance (response time)	TC-09
SR-1	Wake-Word Gating (Security)	TC-10

14. Test Cases

A total of 10 test cases are defined below, covering functional requirements (TC-01 to TC-08), a non-functional performance requirement (TC-09), and a dedicated security validation case (TC-10), satisfying the minimum of 7–10 test cases required for this deliverable.

TC-01: Verify the system enters listening state on wake word.

Test Case ID	TC-01
Requirement Traced	FR-1
Type	Functional
Objective	Verify the system enters listening state on wake word.
Preconditions	System powered on and idle; microphone connected.
Test Steps	1) Speak the wake word clearly. 2) Observe system state/indicator.
Expected Result	System enters active listening state within 1 second of wake word.

TC-02: Verify the Play command starts media playback.

Test Case ID	TC-02
Requirement Traced	FR-2
Type	Functional
Objective	Verify the Play command starts media playback.
Preconditions	The system is listening; the media player is idle/stopped.
Test Steps	1) Say wake word. 2) Say “Play”.
Expected Result	Media playback starts within 2 seconds.

TC-03: Verify the Pause command pauses playback.

Test Case ID	TC-03
Requirement Traced	FR-3
Type	Functional
Objective	Verify the Pause command pauses playback.
Preconditions	Media is currently playing.
Test Steps	1) Say wake word. 2) Say “Pause”.
Expected Result	Playback pauses within 2 seconds.

TC-04: Verify the Resume command resumes playback.

Test Case ID	TC-04
Requirement Traced	FR-4
Type	Functional
Objective	Verify the Resume command resumes playback.
Preconditions	Media is currently paused.
Test Steps	1) Say wake word. 2) Say “Resume”.
Expected Result	Playback resumes within 2 seconds.

TC-05: Verify the Next command skips to next track.

Test Case ID	TC-05
Requirement Traced	FR-5
Type	Functional
Objective	Verify the Next command skips to the next track.
Preconditions	Media is playing with the following track available.
Test Steps	1) Say wake word. 2) Say “Next”.
Expected Result	Player advances to the next track within 2 seconds.

TC-06: Verify the Previous command returns to prior track.

Test Case ID	TC-06
Requirement Traced	FR-6
Type	Functional
Objective	Verify the Previous command returns to prior track.
Preconditions	Media is playing with a prior track available.
Test Steps	1) Say wake word. 2) Say “Previous”.
Expected Result	Player returns to the previous track within 2 seconds.

TC-07: Verify that the Volume Up command increases volume.

Test Case ID	TC-07
Requirement Traced	FR-7
Type	Functional
Objective	Verify that the Volume Up command increases volume.
Preconditions	Media is playing at a known volume level.
Test Steps	1) Say wake word. 2) Say “Volume Up”.
Expected Result	Volume increases by 10% within 1 second.

TC-08: Verify that the Volume Down command decreases volume.

Test Case ID	TC-08
Requirement Traced	FR-8
Type	Functional
Objective	Verify the Volume Down command decreases volume.
Preconditions	Media is playing at a known volume level.
Test Steps	1) Say wake word. 2) Say “Volume Down”.
Expected Result	Volume decreases by 10% within 1 second.

TC-09: Verify command response time under normal noise conditions.

Test Case ID	TC-09
Requirement Traced	NFR-1
Type	Non-Functional (Performance)
Objective	Verify command response time under normal noise conditions.
Preconditions	System idle, quiet room (baseline noise level).
Test Steps	1) Issue each of the 7 commands in turn. 2) Measure time from end of utterance to action execution.
Expected Result	Average response time across all commands is under 2 seconds.

TC-10: Verify that the system does not act on audio without the wake word.

Test Case ID	TC-10
Requirement Traced	SR-1
Type	Security
Objective	Verify that the system does not act on audio without the wake word.
Preconditions	System idle and listening; background audio/TV playing nearby.
Test Steps	1) Play unrelated background speech containing command-like words but no wake word. 2) Observe system behaviour.
Expected Result	No command is executed; system remains idle (confirms wake-word gating, SR-1).